SATHYABAMA
INSTITUTE OF SCIENCE AND TECHNOLOGY
(DEEMED TO BE UNIVERSITY)

TITLE PAGE

- Team Name: ACE
- Theme: AI & MACHINE LEARNING
- PS Category: Software
- Team ID: DEX26271

DEXATHON

2026

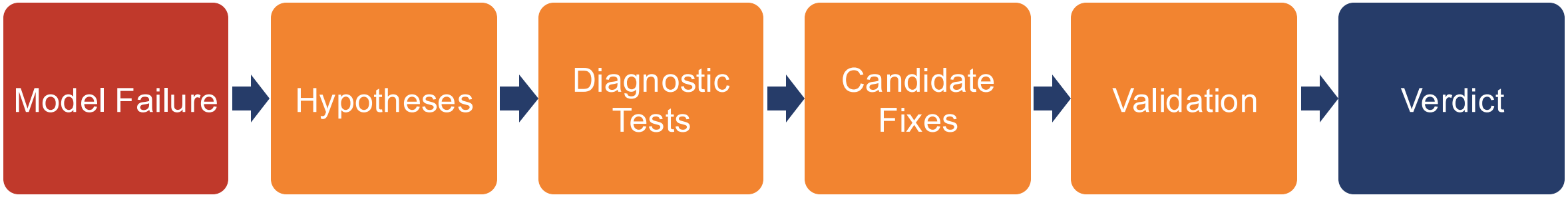
PROBLEM STATEMENT

- AI fails when data changes ,every fix needs proof.
- Finding the failure doesn't reveal the cause.
- A fix can break something else.
- Every fix needs proof before we trust it.

OUR SOLUTION

- Detect model failures.
- Generate competing explanations.
- Challenge explanations through targeted tests.
- Evaluate possible corrections.
- Approve a fix or abstain if evidence is insufficient.

HOW IT WORKS



The original model remains unchanged until a correction passes predefined checks.

AI / ML TECHNOLOGY

- Python, NumPy and pandas
- Scikit-learn for model training and evaluation
- Statistical drift detection
- Hypothesis testing and information-gain-based test selection
- Streamlit for the interactive interface

These are proposed technologies; the implementation is yet to be built.

INNOVATION AND UNIQUENESS

- Combines failure investigation with evidence-based correction testing.
- Selects informative tests within a limited budget.
- Allows the system to reject uncertain solutions.
- Tests whether this strategy outperforms simpler approaches.

IMPORTANCE AND IMPACT

- Gives clear visibility into why AI models fall and what caused it .
- Reduces fixes that don’t actually work .
- Makes model updates safer , because every fix is tested .
- Could support applications in finance, cybersecurity, healthcare research and industrial monitoring.

These are potential benefits to evaluate, not guaranteed outcomes.

EXPECTED OUTCOME AND FUTURE SCOPE

- A working ML failure-investigation prototype.
- A comparison of different diagnostic strategies.
- Measured repair success, false approvals and compute cost.
- Future scope: larger datasets, more model types and controlled deployment.

“An AI should not merely propose a fix. It should provide evidence that the fix deserves to be trusted.”

GitHub Repository Link:
https://github.com/kamalesh0213/My-College-Presentation/blob/main/ppt%20original%2001.pdf